AI-Based Intelligent Border Video Analytics Platform

Technical Stack & Working Explained

SIH 2026 | Problem Statement 26187 | Team Vision Vigilants

IS DOCUMENT KE BAARE MEIN

Yeh document aapke project ke architecture diagram ka har block kholkar samjhata hai: andar actually kya chal raha hai, kaunsi file kya karti hai, aur data camera se React dashboard tak kaise pahunchta hai. Har component par ek status label hai - IMPLEMENTED, PARTIAL ya PLANNED - jo asli code dekhkar lagaya gaya hai, diagram dekhkar nahi.

STATUS LABELS KA MATLAB

IMPLEMENTED	Code mein bana hua hai aur chal raha hai. Demo ho sakta hai.
PARTIAL	Kuch hissa bana hai, kuch nahi. Neeche exactly likha hai ki kya kya.
PLANNED	Abhi nahi bana. Diagram mein hai ya roadmap mein hai. Demo nahi ho sakta.

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SECTION 0

Poora System Ek Nazar Mein

Sabse pehle ek baat clear kar lo, kyunki judge yahin se shuru karta hai. Hamara system ek **chain** hai, parallel boxes nahi. Ek frame aata hai aur woh line se har stage se guzarta hai. Har stage agle stage ko thoda aur matlab deti hai.

Architecture diagram mein saat neele box side by side dikhte hain, jaisa lage ki sab ek saath chal rahe hain. Code mein aisa nahi hai. Order fix hai, aur order hi design hai. Pehle detect, phir ID, phir pose, phir rules, phir score. Agar order badal do to system kaam hi nahi karega, kyunki har stage pichli stage ka output use karti hai.

Ek line mein: **hum recording ko intelligence mein badalte hain, aur hardware ka kharcha zero rakhte hain.**



PERFORMANCE KA DESIGN DECISION

Performance ka ek design decision yahin bata dena achha rehta hai: har frame par poora AI nahi chalta. Har doosra frame analyse hota hai, baaki frame sirf dikhaya jaata hai. Isse operator ko video smooth dikhti hai aur machine par load aadha ho jaata hai. Yeh slide 4 wala *adaptive frame sampling* hai.

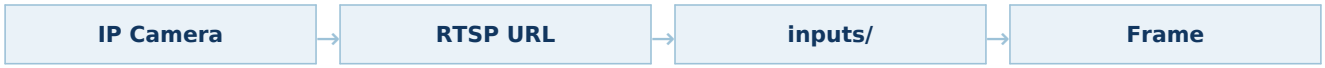
SECTION 1

Video Input Sources

PARTIAL

System ka pehla sawaal yeh hai: video aayegi kahan se. Hamara poora problem statement isi par tika hai ki **naya camera nahi lagana**. Border par pehle se hazaaron CCTV lage hue hain, unhi ko use karna hai.

Iske liye hum camera se seedha baat nahi karte. Hum ek **stream** lete hain. Har IP camera aur har NVR ya DVR RTSP naam ka standard protocol deta hai. Matlab camera ka brand Hikvision ho, CP Plus ho ya Dahua ho, sabse ek hi tareeke se video milti hai.



Hamare project mein

Video file source	IMPLEMENTED	inputs/video_source.py. Abhi chaaron demo camera isi par chal rahe hain.
Webcam source	IMPLEMENTED	inputs/webcam_source.py. Laptop camera se live demo ke liye.
RTSP source	PARTIAL	inputs/rtsp_source.py bana hua hai aur config/cameras.yaml rtsp type ko validate karta hai, lekin abhi tak kisi asli IP camera par test nahi hua. Code ready hai, hardware par try nahi kiya.

JUDGE KO YEH BOLNA

Judge se bolna: **hum camera nahi maangte, hum stream maangte hain.** Isi ek line mein aapka Hardware-Agnostic USP aa jaata hai.

SECTION 2

OpenCV

IMPLEMENTED

OpenCV camera aur AI ke beech ka pul hai. Video apne aap mein AI ke kisi kaam ki nahi hai, kyunki AI model ek **image** samajhta hai, video nahi. OpenCV video ko frame frame karke todta hai.

Uske baad OpenCV hi frame ko model ke layak banata hai: size theek karna, colour format theek karna, aur aakhir mein result wapas frame par draw karna - box, label, zone ki line, track ka rasta.



Hamare project mein

Frame reading	IMPLEMENTED	inputs/ ke teenon source cv2.VideoCapture use karte hain.
Pre-processing	IMPLEMENTED	Resize aur colour handling detection se pehle hoti hai.
Drawing / annotation	IMPLEMENTED	core/annotator.py. Yahi cv2.rectangle se box, cv2.putText se label, cv2.polylines se zone aur cv2.line se boundary draw karta hai. Jo annotated frame dashboard par dikhta hai woh yahin banta hai.

JUDGE KO YEH BOLNA

Ek chhoti si baat jo achhi lagti hai: jo annotated frame operator dekhta hai, wahi frame evidence mein bhi save hota hai. Matlab baad mein dekhne wale ko pata chalta hai ki system ne kya dekha tha aur kyun.

SECTION 3

YOLO Object Detection

IMPLEMENTED

YOLO ka poora naam hai You Only Look Once. Naam ka matlab yahi hai ki woh image ko ek hi baar dekhkar bata deta hai ki usmein kya kya hai aur kahan hai. Purane tareeke image ko tukdon mein baar baar scan karte the, isliye woh dheeme the. YOLO ek shot mein kaam karta hai, isliye live video par chal sakta hai.

Har detection ke saath do cheezein aati hain: ek **bounding box** (object kahan hai) aur ek **confidence score** (model kitna sure hai). Confidence kam ho to hum us detection ko chhod dete hain.

```
Frame #248 par YOLO ka output:

  person   conf 0.91   box (412, 180, 480, 395)
  car      conf 0.87   box (120, 300, 340, 470)

Matlab: is frame mein ek person aur ek vehicle detect hua.
```

Hamare project mein

YOLOv8n	IMPLEMENTED	models/yolov8n.pt. 'n' matlab nano, sabse chhota aur tez version, taaki bina GPU ke bhi chale.
Sirf 6 class	IMPLEMENTED	YOLO 80 cheezein pehchaan sakta hai. Hum sirf 6 le rahe hain: person, car, motorcycle, bus, truck, backpack. Baaki 75 classes border par shor hain. Isse speed bhi badhti hai aur false alarm bhi ghatta hai.

JUDGE KO YEH BOLNA

Judge poochh sakta hai ki aapne YOLO train kiya kya. Sach bolna: **humne YOLO train nahi kiya**, pretrained weights use kiye hain. Hamara kaam uske upar ka hai. Iska detail section 21 mein hai.

SECTION 4

Tracking (ByteTrack)

IMPLEMENTED

Sirf detection kaafi nahi hai, aur iski wajah samajhna zaroori hai. YOLO har frame ko naya maanta hai. Frame 100 mein use ek aadmi dikhta hai aur frame 101 mein bhi aadmi dikhta hai, lekin YOLO ko yeh nahi pata ki dono **ek hi** aadmi hain.

Bina tracking ke aap kuch bhi nahi keh sakte. Na speed, na direction, na 'yeh aadmi 40 second se yahin khada hai'. Tracking har aadmi ko ek ID deti hai aur woh ID frame dar frame usi ke saath rehti hai.

```
ByteTrack ke baad:

  Person #01   frame 100 -> 101 -> 102 ...   (wahi banda)
  Person #02   frame 104 se shuru
  Vehicle #07  frame 110 se shuru

Ab hum keh sakte hain: Person #02 pichle 12 second se fence ke paas hai.
```

Hamare project mein

ByteTrack	IMPLEMENTED	Ultralytics ke model.track() ke andar hi chalta hai, matlab detection aur tracking ek hi call mein ho jaate hain.
Apni confirmation layer	IMPLEMENTED	core/tracker.py. Yeh hamari apni layer hai ByteTrack ke upar. Teen niyam: track kam se kam 3 baar dikhe tabhi ginti mein aata hai, detector ke 5 frame miss karne par bhi track zinda rehta hai, aur 30 frame tak na dikhe to retire ho jaata hai.

JUDGE KO YEH BOLNA

Yeh confirmation layer hamara **false alarm ka pehla bachaav** hai. Ek frame ki jhalak kabhi alert nahi ban sakti, kyunki ek frame ka track ginti mein hi nahi aata.

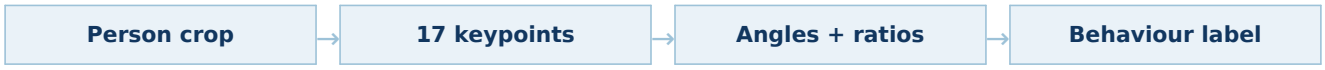
SECTION 5

Pose Estimation

IMPLEMENTED

Bounding box sirf itna batata hai ki aadmi kahan hai. Woh yeh nahi batata ki aadmi **kar kya raha hai**. Khada hua aadmi aur jhuka hua aadmi, dono ka box lagbhag ek jaisa dikhta hai.

Pose estimation aadmi ke sharir par 17 point nikaalta hai: sar, dono kandhe, kohni, kalai, kamar, ghutne, takhne. In points ke aapas ke angle aur doori se pata chalta hai ki aadmi seedha khada hai, jhuka hua hai, ya haath upar karke kuch chadh raha hai.



Hamare project mein

YOLOv8n-pose	IMPLEMENTED	models/yolov8n-pose.pt, COCO ke 17 keypoints.
Sirf zaroorat par	IMPLEMENTED	Pose har aadmi par nahi chalta. Sirf confirmed track par, har doosre analysed frame par, aur sirf un logon par jo frame mein kaafi bade dikh rahe hain. Door ka chhota figure chhod diya jaata hai kyunki uske keypoints waise bhi bharose ke laayak nahi hote.
Pose history	IMPLEMENTED	core/pose_history.py. Ek frame ka pose kaafi nahi hai, isliye hum kuch frames ka pose jama karke phir faisla lete hain.

JUDGE KO YEH BOLNA

Imaandaari se ek baat: CROUCHING aur JUMPING ka code bana hua hai aur test bhi paas karta hai, lekin hamari demo footage mein yeh harkatein hoti hi nahi, isliye demo mein yeh label nahi dikhega. Judge ko pehle hi bata dena behtar hai.

SECTION 6

Behaviour Analysis

IMPLEMENTED

Yahan detection ek **activity** ban jaati hai. Akela detection bekaar hai. Detection plus movement plus pose milkar ek matlab wali baat banti hai.

Sabse zaroori design decision yahin hai, aur yahi hamara sabse majboot innovation point hai. Hum speed aur doori **pixel mein nahi naapte**. Hum har aadmi ki apni body height ke hisaab se naapte hain.

Wajah simple hai. Har camera alag height aur alag angle par lagta hai, isliye ek hi aadmi ek camera mein bada aur doosre mein chhota dikhta hai. Agar pixel mein naapenge to har camera ke liye alag calibration karni padegi, jo hazaaron camera ke liye namumkin hai. Body height se naapne par number har camera par ek jaisa rehta hai.

```
Formula ki jagah scale-free measurement:

  walking  =  ~0.45 body-height / second
  running  =  ~1.90 body-height / second

Person detected + movement pattern + pose
        =  WALKING / RUNNING / CROUCHING / CLIMBING
```

Hamare project mein

Behaviour classifier	IMPLEMENTED	core/behaviour_classifier.py
Movement + loitering	IMPLEMENTED	core/behaviour_engine.py, core/movement.py
Boundary interaction	IMPLEMENTED	core/boundary_interaction.py. Fence ke saath rishta ek ladder hai: NONE, APPROACHING, AT_BOUNDARY, INTERACTING, CROSSED. System pehle sirf note karta hai, alert turant nahi deta.

JUDGE KO YEH BOLNA

Naye camera par koi setup, koi calibration, koi measurement nahi chahiye. Stream jodo aur system chal padta hai. **Yeh line judge ke 'isme naya kya hai' wale sawaal ka sabse achha jawaab hai.**

SECTION 7

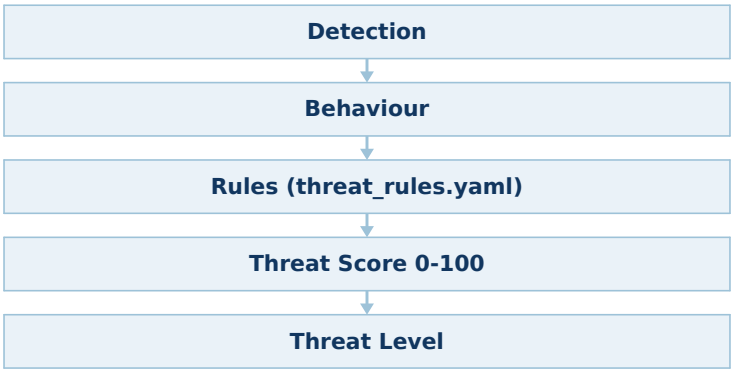
Threat Engine

IMPLEMENTED

Yeh hamare project ka dil hai, isliye ise sabse saaf bolna.

YOLO object **detect** karta hai. YOLO yeh nahi batata ki woh object khatarnak hai ya nahi. YOLO ke liye fence ke paas raat ko jhuka hua aadmi aur din mein road par chalta aadmi, dono bas 'person' hain. Khatre ka faisla YOLO nahi, **Threat Engine** karta hai.

Threat Engine ek hi detection nahi dekhta. Woh us aadmi ka poora context dekhta hai: kahan hai, kis time hai, kaise chal raha hai, kitni der se hai, akela hai ya group mein. Har factor ka apna weight hai. Sab weight jodkar 0 se 100 ke beech ek score banta hai.




```
LEVEL THRESHOLDS
```

```
  0  - 24   LOW        sirf log hota hai, alert nahi
  25 - 49   MEDIUM     dashboard par dikhta hai
  50 - 74   HIGH       alert banti hai
  75 - 100  CRITICAL   sabse upar, turant dikhta hai

UDAHARAN

  Person + Walking + Normal area          ->  LOW
  Person + Crouching + Restricted zone    ->  MEDIUM / HIGH
  Person + Restricted + Night + Unusual   ->  HIGH / CRITICAL
```

Hamare project mein

Threat Engine	IMPLEMENTED	core/threat_engine.py. 16 alag alag factor, har ek ka weight config/threat_rules.yaml se aata hai.
Pure function design	IMPLEMENTED	Threat Engine apne andar kuch yaad nahi rakhta. Usko context do, woh score wapas karta hai. Isse test karna asaan hai aur nateeja hamesha ek jaisa rehta hai.

JUDGE KO YEH BOLNA

Ek design choice zaroor batana: **rules code ke andar nahi, ek YAML file mein hain**. Matlab jis sector mein raat ki harkat aam baat hai, wahan ka officer bina programmer ke us factor ka weight badal sakta hai. Field mein yeh bahut kaam aata hai.

SECTION 8

Rules & Configuration

IMPLEMENTED

Hamara poora system config files se chalta hai, hardcoded numbers se nahi. Iska faayda yeh hai ki har border sector ki apni zaroorat alag hoti hai, aur unke liye alag software nahi banana padta.

```
config/ folder

  cameras.yaml        kaunsa camera, kaunsa source, kaunsa zone profile
  zones.yaml          restricted zone ka polygon aur boundary line
  threat_rules.yaml   16 factor aur unke weights
  application.yaml    frame stride, pose interval, ANPR cooldown, etc.

RULE KE UDAHARAN

  restricted zone intrusion      person ek mana kiye gaye polygon mein
  night-time movement            raat ke ghanton mein harkat
  crouching / climbing           pose se mila behaviour
  loitering                      ek hi jagah par lambe samay tak
  wrong-direction crossing       boundary line galat taraf se paar
  vehicle in restricted area     gaadi mana kiye gaye area mein
```

Hamare project mein

YAML driven rules	IMPLEMENTED	config/threat_rules.yaml, 16 factor.
Per-camera zone profile	IMPLEMENTED	config/cameras.yaml mein har camera apna zone profile chunta hai. Jis camera ko zone nahi chahiye woh 'none' rakh sakta hai.

SECTION 9

Event Correlation & Threat Scoring

IMPLEMENTED

Agar hum har detection ko alag alag treat karein to do dikkat aati hain. Ek, operator ko ek hi ghatna ke liye paanch alag card dikhte hain. Do, asli khatra chhup jaata hai kyunki akela akela har cheez chhoti lagti hai.

Correlation ka matlab hai ki ek hi track ki alag alag baatein jodkar ek kahani banti hai, aur us poori kahani par ek score lagta hai.

```
ALAG ALAG DEKHEIN TO

  person detected          -> normal
  zone entry               -> chhoti baat
  crouching                -> ho sakta hai
  raat ka time             -> normal

JODKAR DEKHEIN TO

  Person #02  +  restricted zone  +  crouching  +  22:41 raat
        =  HIGH RISK INCIDENT, ek hi card, saare reasons ke saath
```

Hamare project mein

Incident Manager	IMPLEMENTED	core/incident_manager.py. Ek dedupe window rakhta hai, taaki ek hi ghatna baar baar naya incident na bane.
Order matters	IMPLEMENTED	Pipeline mein zone incident pehle raise hota hai aur behaviour incident baad mein, jaan bujhke. Isse jo aadmi restricted zone mein bhi hai aur crouch bhi kar raha hai, uske liye **ek** card banta hai, do nahi.

SECTION 10

Alerts & Notifications

PARTIAL

Jab incident ka threat level alert ke layak hota hai, tab Alert Manager ek alert banata hai. Alert mein sirf 'khatra hai' nahi hota, usmein poori wajah hoti hai, taaki operator bina video dekhe hi samajh jaye ki dekhna kyun hai.

```
ALERT KA SHAKL

  HIGH THREAT
  Camera    : CAM-03
  Track     : Person #02
  Behaviour : Crouching
  Reason    : Unauthorized movement in restricted zone
  Time      : 22:41:08
```

Hamare project mein

Alert Manager	IMPLEMENTED	core/alert_manager.py. Incident se alert banata hai, severity nikaalta hai, aur acknowledge / resolve bhi sambhalta hai.
Dashboard par alert	IMPLEMENTED	React dashboard ke LIVE THREAT EVENTS panel mein aata hai, /api/events se. Camera ke alert counter bhi live badalte hain.
Email / SMS / webhook	PLANNED	Alert Manager mein NotificationChannel ka abstraction bana hua hai, matlab naya channel jodna asaan hai. Lekin abhi **koi email, SMS ya webhook channel likha nahi gaya**. Sirf in-app channel hai.

JUDGE KO YEH BOLNA

Yeh saaf bolna: **notification abhi dashboard tak hai, phone tak nahi.** Architecture taiyaar hai, channel likhna baaki hai.

SECTION 11

Evidence Storage

PARTIAL

Alert aa gayi, operator ne dekh liya, aur baat khatam. Lekin asli kaam baad mein shuru hota hai. Enquiry mein, court mein, ya senior officer ke saamne sabse pehla sawaal yahi hota hai: **proof kahan hai**.

Isliye jab bhi incident banta hai, system us waqt ki tasveer apne aap save kar leta hai. Operator ko kuch karna nahi padta, aur ghanton ki recording mein peeche jaakar dhoondhna nahi padta.

Hamare project mein

Annotated full frame	IMPLEMENTED	core/evidence.py. Poora frame, overlay ke saath. Matlab usmein box, label aur zone bhi dikhte hain, to baad mein dekhne wale ko pata chalta hai ki system ne kya dekha tha.
Object crop	IMPLEMENTED	Us aadmi ya gaadi ka close-up crop.
Plate crop	IMPLEMENTED	ANPR ka plate crop alag se save hota hai.
Video clip	PLANNED	Architecture diagram mein likha hai 'Images / Video / Logs', lekin **video clip save karne ka code abhi nahi hai**. Sirf JPEG image save hoti hain. Yeh saaf bolna hai.
Logs	IMPLEMENTED	system_logs aur audit_logs, dono database mein hain.

JUDGE KO YEH BOLNA

Evidence ka path aur uski entry database mein jaati hai, aur dashboard us image ko /api/evidence se dikhata hai. Matlab alert par click karke operator turant tasveer dekh sakta hai.

SECTION 12

Database (SQLite)

IMPLEMENTED

SQLite ek file-based database hai. Iska matlab hai ki alag se koi database server install nahi karna padta, poora database ek file hai. Border outpost ke liye yeh bilkul sahi hai, kyunki wahan IT team nahi baithi hoti jo MySQL server sambhale.

Database ke bina system ki yaadasht nahi hoti. Har alert screen se gayab ho jaati, aur 'pichle teen mahine mein kis sector mein sabse zyada harkat hui' jaisa sawaal kabhi jawaab nahi milta.

```
data/database/borderguard.db  mein 17 table

  cameras        camera ki config aur aakhri status
  sessions       kaunsa camera kab se kab tak chala
  tracks         har track ka record
  detections     sampled detection rows
  zones          per camera zone geometry
  incidents      banaye gaye incident
  alerts         raise ki gayi alert
  evidence       evidence file ka path aur metadata
  plates         ANPR se padhi gayi number plate
  watchlist      dhyaan rakhne wali plate
  system_logs    system ke log
  settings       runtime settings
  users          login accounts (password hash)
  auth_sessions  login session
  audit_logs     kisne kya kiya
  schema_version, sqlite_sequence
```

Hamare project mein

Poora schema IMPLEMENTED database/ package, 17 table, migrations ke saath.

JUDGE KO YEH BOLNA

Judge poochh sakta hai ki scale par SQLite chalega kya. Sach: ek BOP ke liye bilkul chalega. Kayi BOP ka central system banega to PostgreSQL par jaana hoga, aur kyunki hamari database layer alag hai, badalna mushkil nahi hai.

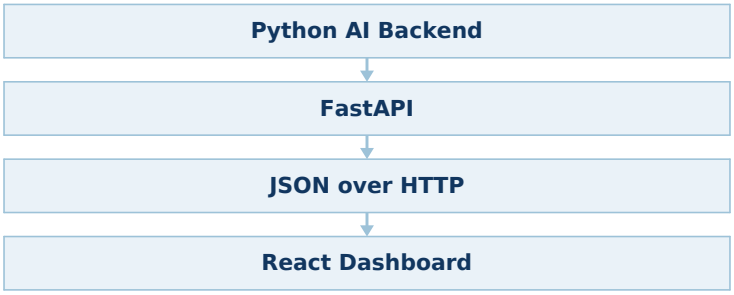
SECTION 13

FastAPI

IMPLEMENTED

AI Python mein chal raha hai aur dashboard browser mein React par chal raha hai. Dono alag duniya hain. Beech mein ek pul chahiye jo Python ka data browser tak pahuncha sake. Woh pul FastAPI hai.

FastAPI ek web framework hai jo Python ke data ko **JSON** mein badalkar HTTP par bhej deta hai. React us JSON ko padhkar screen par dikha deta hai.




```
GET /api/events ka jawaab (chhota sa namoona)

  {
    "events": [
```

```
      {
        "camera_id"    : "CAM-03",
        "track_number" : 2,
        "threat_level" : "HIGH",
        "behaviour"    : "CROUCHING",
        "headline"     : "Unauthorized movement in restricted zone",
        "ts"           : 1757800868.42,
        "evidence_id"  : 118
      }
    ]
  }
```


Hamare project mein

FastAPI backend	IMPLEMENTED	api/server.py, 18 endpoints.
Single process design	IMPLEMENTED	FastAPI hi React ka build (frontend/dist) bhi serve karta hai. Matlab do server chalane ki zaroorat nahi, sirf port 8000. Isi wajah se CORS ki dikkat bhi nahi aati.
MJPEG video stream	IMPLEMENTED	/api/cameras/{id}/stream se annotated live video browser tak jaati hai.

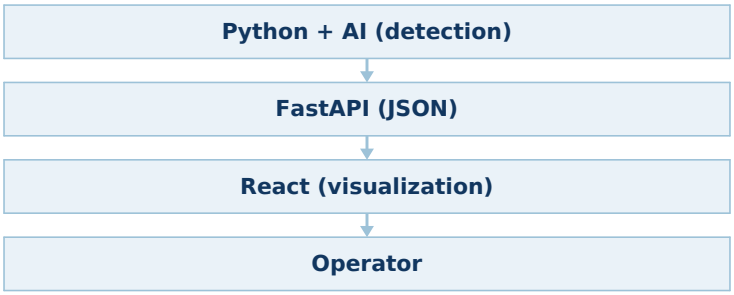
SECTION 14

React Dashboard

PARTIAL

Yeh sabse zaroori galatfehmi door karni hai. **React koi AI nahi karta.** React mein YOLO nahi chalta, detection nahi hoti, tracking nahi hoti. React sirf dikhata hai aur control deta hai.

Saara AI Python mein hota hai. React sirf FastAPI se JSON leta hai aur usko sundar tareeke se screen par sajaata hai.



Hamare project mein

Dashboard (command wall)	IMPLEMENTED	Live camera feeds, camera status, live counters, ANPR plates panel, aur LIVE THREAT EVENTS - sab ek screen par. Saara data asli backend se.
Camera Management	IMPLEMENTED	Camera start / stop aur per-camera detail.
Detection Logs	IMPLEMENTED	/api/events se poora recorded event log, per camera ya sab camera.
Alerts page	PARTIAL	Abhi sirf live camera stats se counter dikhata hai. Purani alert ki history yeh page nahi laata. Asli event list Dashboard aur Detection Logs par hai.

Analytics page	PARTIAL	Abhi sirf live totals: persons, vehicles, detections, active cameras. Koi trend chart ya historical analysis nahi hai.
User Management + Audit Logs	IMPLEMENTED	Do role, aur 18 tarah ke audit action ka log.
Reports / export	PLANNED	Diagram mein 'Reports' likha hai, lekin abhi koi report banane ya export karne ka feature nahi hai.

JUDGE KO YEH BOLNA

Ek aur cheez jo saaf rakhni hai: permission React mein nahi, **backend mein** check hoti hai. Agar koi browser ka code badal bhi de to bhi API use data nahi degi.

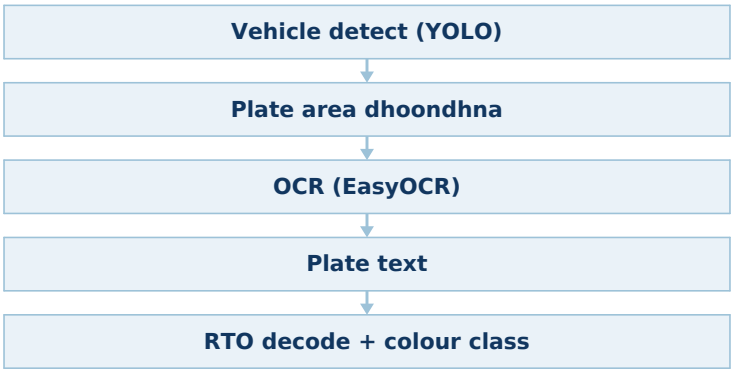
SECTION 15

ANPR (Number Plate Recognition)

IMPLEMENTED

ANPR ka matlab hai gaadi ka number plate apne aap padh lena. Border checkpoint par yeh kaam ka hai kyunki gaadi ka number ek pakka record hota hai, chehre ki tarah dhundhla nahi.

Hamara ANPR sirf plate padhkar nahi rukta. Padhne ke baad woh do aur kaam karta hai jo asli value dete hain.




```
EK ASLI READING

  raw OCR      HR26800008
  grammar fix  HR26BO0008      (0 -> O series letter position par)
  RTO decode   HR = Haryana,  HR26 = Gurgaon
  plate colour safed  ->  Private vehicle

  peela plate  ->  Commercial / Taxi
  hara plate   ->  Electric vehicle
```

Hamare project mein

OCR engine	IMPLEMENTED	EasyOCR 1.7.2 project ke .venv mein install hai, isliye ANPR abhi chal raha hai. core/anpr.py mein Tesseract ka backend bhi hai as a fallback.
RTO decode	IMPLEMENTED	core/rto.py. 41 state code aur 998 district code ki table. Plate se state aur district nikaalta hai.
Plate colour class	IMPLEMENTED	core/plate_colour.py. Plate ke rang se gaadi ka type nikaalta hai, bina kisi bahar ke database ke.

Grammar repair	IMPLEMENTED	core/plate_grammar.py. OCR ki aam galtiyan theek karta hai, jaise 0 aur O. **State ke do akshar kabhi repair nahi karta**, kyunki ek baar repair ne Haryana ki gaadi ko Odisha ka bana diya tha. Uska regression test bhi likha hua hai.

JUDGE KO YEH BOLNA

Accuracy ki asli limit saaf bolna. Hamare test clip par lagbhag 20 gaadiyon mein se 5 se 8 ki plate padh paayi. Door ki, tedhi, dhundhli ya motion blur wali plate chhoot jaati hai. Jo padh jaata hai woh saaf record hota hai; jo nahi padhta, uske liye system andaaza nahi lagata.

SECTION 16

Anomaly Detection (Isolation Forest)

IMPLEMENTED

Rules ki ek seema hai: rule sirf wahi pakad sakta hai jo humne pehle soch rakha ho. Lekin border par aisi harkat bhi ho sakti hai jo humne socha hi nahi.

Isolation Forest is dikkat ka jawaab hai. Isko yeh batana nahi padta ki galat kya hai. Yeh khud dekh leta hai ki is camera par aam tor par log kaise chalte hain, aur jab koi us pattern se hatkar chalta hai to usko 'anomalous' mark kar deta hai.

```
HAR TRACK KE KHATM HONE PAR

  us track ke features nikalkar model ko sample ke taur par diye jaate hain
  25 samples jama hone par model apne aap fit ho jaata hai
  uske baad live tracks ko score milta hai: NORMAL ya ANOMALOUS

  Normal     : log aate jaate hain, ek jaisi raftaar, ek jaisa rasta
  Anomalous  : koi ulti disha, ruk ruk kar, ya bilkul alag pattern
```

Hamare project mein

Isolation Forest	IMPLEMENTED	core/anomaly_engine.py, scikit-learn ka IsolationForest. Model models/behaviour_anomaly.joblib mein save hota hai.
Per-camera learning	IMPLEMENTED	Model us camera ke apne data se seekhta hai. Matlab har camera ka 'normal' alag ho sakta hai, aur yeh sahi bhi hai.

JUDGE KO YEH BOLNA

Yeh aapka **asli 'machine learning' wala jawaab** hai. Jab judge poochhe ki aapne kaunsa model train kiya, to yeh bolna: YOLO pretrained hai, lekin anomaly model hum har camera par live train karte hain.

SECTION 17

Edge + Cloud Hybrid

PARTIAL

Border ka sabse bada sach yeh hai ki wahan internet bharosemand nahi hota. Isliye AI ko cloud par rakhna practical nahi hai. Agar har frame cloud bhejenge to bandwidth bhi khatam aur alert bhi der se aayegi.

Iska jawaab edge computing hai. AI wahin chalega jahan camera hai, BOP par hi. Alert wahin banegi. Cloud sirf summary ke liye use hoga, video ke liye nahi.

```
EDGE (BOP par)                    CLOUD (headquarters)

  Camera                            Central monitoring
  Local AI (YOLO, pose)             Long-term storage
  Threat detection                  Reports
  Turant alert                      Multi-site analytics

EDGE KA FAAYDA
  low latency, kam bandwidth, internet na ho to bhi kaam,
  aur privacy behtar kyunki video bahar nahi jaati
```

Hamare project mein

Edge processing	IMPLEMENTED	Hamara poora system abhi ek hi machine par ek hi process mein chalta hai. **Yahi edge deployment ka shape hai.** Demo mein aapka Mac hi edge device hai.
Cloud tier	PLANNED	Central server, multi-BOP aggregation, long-term storage aur multi-site analytics - **inmein se kuch bhi abhi nahi bana hai**. Architecture modular hai isliye jodna mushkil nahi, par abhi hai nahi.

JUDGE KO YEH BOLNA

Bolne ka sahi tareeka: **edge wala hissa aaj chal raha hai, cloud wala hissa design mein hai.** Yeh kehna ki hybrid poora bana hua hai, galat hoga.

SECTION 18

Scalability & Modular Architecture

IMPLEMENTED

Modular ka matlab hai ki har kaam apni alag file mein hai, aur ek file badalne se doosri nahi tootti. Yeh sunne mein choti baat lagti hai, lekin asli project mein yahi sabse zyada kaam aata hai.

Jaise, ANPR ko humne baad mein joda. Poora system dobara nahi likhna pada, kyunki ANPR apni alag file thi aur baaki pipeline usse independent thi. Agar ANPR ka OCR engine install na ho to system 'ANPR unavailable' bolta hai aur baaki sab waise ka waisa chalta rehta hai.

```
core/ ke independent module

  detector.py        YOLO          badalna ho to naya model plug karo
  tracker.py         ByteTrack     tracking algorithm alag se badle
  pose.py            pose model
  anpr.py            ANPR          OCR engine swap ho sakta hai
  threat_engine.py   rules         YAML se, code chhue bina
  anomaly_engine.py  ML
  evidence.py        storage
  database/          DB layer      SQLite se PostgreSQL par jaa sakte hain
```

Hamare project mein

Independent modules	IMPLEMENTED	Har module apni file mein, aur 491 automated test har build par chalte hain taaki ek module badalne se doosra chupke se na toote.

Microservices	PLANNED	Diagram ke 'Scalable & Modular' box mein 'Microservices Architecture' likha hai. Sach yeh hai ki abhi **ek hi FastAPI process** hai. Python packages modular hain, lekin alag alag service nahi hain. Yeh saaf bol dena behtar hai.

SECTION 19

Blockchain

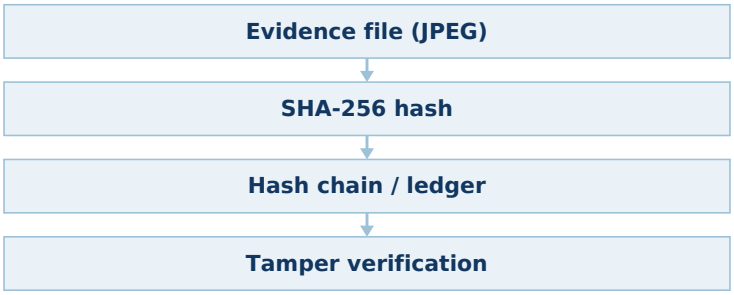
PLANNED

Architecture diagram ke neeche likha hai 'A Multi-Layered AI & Blockchain Framework', aur theme bhi Blockchain & Cybersecurity hai. Isliye is par jawaab tayyar hona chahiye.

Seedhi baat: **project ke code mein blockchain nahi hai.** Na hash chain, na ledger, kuch nahi. Isko implemented bolna galat hoga aur pakde jaane par bharosa toot jaayega.

Lekin theme ka doosra hissa, cybersecurity, asli mein bana hua hai, aur woh kaafi majboot hai. Password kabhi plain text mein store nahi hota, scrypt se hash hota hai. Login token bhi seedha store nahi hota, uska fingerprint store hota hai. Do role hain aur permission backend mein check hoti hai. Aur 18 tarah ke audit log bante hain, jinmein EVIDENCE_EXPORTED aur ACCESS_DENIED bhi shaamil hai.

Blockchain ki jagah bhi ekdum saaf hai, aur yeh roadmap mein hai.




```
PROPOSED DESIGN (abhi bana nahi hai)

  har evidence file ka SHA-256 hash nikaalo
  har naye record ke andar pichle record ka hash rakho
  ab agar koi beech ka ek record ya ek image badalta hai
  to aage ki poori chain match karna band kar deti hai
  aur chhed chhaad turant pakdi jaati hai
```

Hamare project mein

Blockchain / hash chain	PLANNED	Code mein nahi hai. Roadmap mein hai.
Cybersecurity layer	IMPLEMENTED	core/auth.py: scrypt password hashing, session token fingerprint, hmac.compare_digest, role check backend par, 18 audit action.

JUDGE KO YEH BOLNA

Bolne ka tareeka: **jo bana hai usko confidence se batao, jo nahi bana uska plan batao.** Yeh mat bolna ki blockchain zaroori nahi hai, usse lagta hai aap theme se bhaag rahe ho.

SECTION 20

Complete End-to-End Example

Ab sab kuch ek saath dekhte hain. Yeh ek asli scenario hai, aur agar judge poochhe ki 'YOLO se lekar React tak exactly kya hota hai', to yahi sunana hai.

SCENARIO

Scenario: Raat ke 22:41 baje, CAM-03 (Sector Charlie, border wall) par ek aadmi restricted zone mein ghusta hai aur deewar ke paas jhuk jaata hai.

01 **Frame aata hai**
CAM-03 ki stream se OpenCV ek frame nikaalta hai. Yeh 248th frame hai. Frame stride 2 hai aur 248 even hai, isliye yeh frame analyse hoga.

02 **Scene context**
core/low_light.py naapta hai ki frame kitna andhera hai. Raat hai, to yeh reading 'dark' aati hai. Yeh reading isi frame ke har aage wale step mein use hogi, taaki ek frame kabhi 'lit' aur 'dark' dono na ban jaye.

03 **YOLO detect karta hai**
YOLOv8n bolta hai: person, confidence 0.91, box (412, 180, 480, 395).

04 **ByteTrack ID deta hai**
ByteTrack dekhta hai ki yeh wahi aadmi hai jo pichle frames mein tha, aur usko Person #02 ki ID deta hai. core/tracker.py confirm karta hai kyunki yeh track 3 se zyada baar dikh chuka hai.

05 **Pose chalta hai**
Person #02 frame mein kaafi bada hai, isliye pose estimation chalta hai. 17 keypoints nikalte hain. Ghutne mude hue hain aur kamar neeche hai, to behaviour classifier bolta hai CROUCHING.

06 **Zone rule trigger hota hai**
core/zone_engine.py dekhta hai ki Person #02 ka reference point restricted zone ke polygon ke andar hai. Zone violation event banta hai.

07 **Boundary ladder aage badhti hai**
core/boundary_interaction.py ka state NONE se APPROACHING se AT_BOUNDARY tak pahunch chuka hai. Yaad rahe, yahan abhi alert nahi jaati. Pehle arm hota hai, phir confirm.

08 **Threat Engine score nikaalta hai**
Ab Threat Engine ke paas poora context hai: restricted zone, crouching, raat ka time, boundary ke paas, aur anomaly status. threat_rules.yaml ke weights jodkar score 68 aata hai. 68 matlab HIGH.

09 **Incident banta hai**
core/incident_manager.py dekhta hai ki isi track ka koi incident abhi abhi nahi bana. Naya incident banta hai, aur crouching wala behaviour usi incident mein fold ho jaata hai, alag card nahi banta.

10 **Evidence save hoti hai**
core/evidence.py do JPEG likhta hai: annotated poora frame (box, label aur zone ke saath) aur Person #02 ka close-up crop. Dono ka path database ki evidence table mein jaata hai.

11 **Alert raise hoti hai**
core/alert_manager.py alert banata hai: HIGH THREAT, CAM-03, Person #02, CROUCHING, 'Unauthorized movement in restricted zone', 22:41:08.

12 **Database mein sab likha jaata hai**
incidents, alerts, evidence aur tracks - chaaron table mein entry jaati hai, timestamp ke saath.

13 **FastAPI JSON deta hai**
React ka dashboard har second /api/events aur /api/dashboard poll karta hai. Agli call par yeh naya event JSON mein aa jaata hai.

14 **React dashboard dikhata hai**
LIVE THREAT EVENTS panel mein laal HIGH ka card upar aa jaata hai, poore reason ke saath. Camera ka alert counter badh jaata hai. Card par click karke operator evidence image dekh sakta hai, jo /api/evidence se aati hai.

15 **Operator faisla leta hai**
Operator ne screen nahi dekhi thi. Usne alert dekhi, ek click mein tasveer dekhi, aur teen second mein tay kar liya ki patrol bhejni hai ya nahi. **Yahi poore project ka point hai.**

SECTION 21

Implemented vs Proposed

Yeh table sabse zaroori hai. Ise dekhkar jaana, taaki galti se koi aisi cheez na bol do jo project mein hai hi nahi. Agar judge 'demo dikhao' bole aur cheez mili hi nahi, to usse zyada nuksaan kuch nahi karta.

COMPONENT	STATUS	ROLE / HONEST NOTE
Video file input	IMPLEMENTED	Chaaron demo camera isi par chal rahe hain.
Webcam input	IMPLEMENTED	Live demo ke liye.
RTSP / IP camera input	PARTIAL	Code bana hua hai, asli camera par test nahi hua.
OpenCV frame handling	IMPLEMENTED	Frame nikalna, pre-process, aur annotation.
YOLOv8 object detection	IMPLEMENTED	Pretrained weights, 80 mein se 6 class.
ByteTrack tracking	IMPLEMENTED	Plus apni confirmation layer (3 hits, 5 coast, 30 age).
Pose estimation	IMPLEMENTED	YOLOv8n-pose, 17 keypoints, selective.
Behaviour analysis	IMPLEMENTED	Body-height based, scale-free, per-camera calibration ki zaroorat nahi.
Boundary interaction	IMPLEMENTED	5-state ladder, arm-and-confirm.
Threat Engine	IMPLEMENTED	16 factor, YAML se configurable, 4 level.
Rules & configuration	IMPLEMENTED	4 YAML file, code change ke bina tune hota hai.
Event correlation / dedupe	IMPLEMENTED	Incident Manager, ek ghatna ka ek card.
Anomaly detection	IMPLEMENTED	Isolation Forest, har camera par live seekhta hai.
ANPR + OCR	IMPLEMENTED	EasyOCR 1.7.2 installed. Accuracy limit section 15 mein hai.
RTO state / district decode	IMPLEMENTED	41 state, 998 district code.
Plate colour classification	IMPLEMENTED	Safed, peela, hara se vehicle type.
Evidence (images)	IMPLEMENTED	Annotated frame + crop + plate crop, JPEG.
Evidence (video clips)	PLANNED	Abhi video clip save nahi hota.
SQLite database	IMPLEMENTED	17 table.
FastAPI backend	IMPLEMENTED	18 endpoint, React build bhi yahi serve karta hai.
React dashboard (main)	IMPLEMENTED	Command wall: feeds, status, stats, plates, events.
Alerts page (history)	PARTIAL	Abhi sirf live counters, purani history nahi.
Analytics page	PARTIAL	Abhi sirf live totals, koi trend chart nahi.
Reports / export	PLANNED	Abhi nahi bana.
In-app notification	IMPLEMENTED	Dashboard par alert dikhti hai.
Email / SMS / webhook	PLANNED	Abstraction ready, channel nahi likha.
Auth, roles, audit logs	IMPLEMENTED	scrypt, 2 role, backend check, 18 audit action.

COMPONENT	STATUS	ROLE / HONEST NOTE
Low-light measurement	IMPLEMENTED	Frame kitna andhera hai, yeh naapta hai.
Low-light image enhancement	PLANNED	Enhancement ka code abhi nahi hai.
RGB-Thermal fusion	PLANNED	Thermal ka koi code nahi. Strategy hai, build nahi.
Edge deployment	IMPLEMENTED	Ek machine, ek process. Demo mein Mac hi edge hai.
Cloud tier / multi-BOP	PLANNED	Design mein hai, bana nahi hai.
Microservices	PLANNED	Abhi ek hi FastAPI process hai.
Blockchain / hash chain	PLANNED	Code mein nahi hai. Roadmap mein hai.
Face recognition	PLANNED	Jaan bujhke scope se bahar rakha gaya hai.
491 automated tests	IMPLEMENTED	Har build par chalte hain aur paas hote hain.

SECTION 22

SIH Judge Ko Kaise Explain Karein

Judge ko 60 second mein kaise explain karein

Sir, border par pehle se hazaaron CCTV lage hue hain aur woh 24 ghante recording karte hain. Dikkat camera ki nahi hai, dikkat yeh hai ki us footage ko dekhne wala ek insaan hai, aur koi bhi insaan 20-25 minute se zyada screen par dhyaan nahi de sakta.

Hamara solution IBVAP hai. Hum naya camera nahi lagate. Usi purani CCTV feed ko hum AI se padhte hain. YOLO aadmi aur gaadi detect karta hai, ByteTrack unko ID deta hai, pose estimation batata hai ki woh kaise chal raha hai, aur Threat Engine yeh sab jodkar 0 se 100 ka score nikaalta hai.

Agar score HIGH aata hai to incident banta hai, us waqt ki tasveer apne aap evidence mein save hoti hai, aur FastAPI ke through alert React dashboard par pahunch jaati hai. Matlab jawaan ab camera nahi dekhta, sirf alert dekhta hai.

Ek line mein: **hum recording ko intelligence mein badalte hain, aur hardware ka kharcha zero rakhte hain.**

10 Common Technical Questions

Q1. YOLO aur tracking mein kya farak hai?

YOLO batata hai ki is frame mein kya hai. Tracking batati hai ki pichle frame wala aur is frame wala ek hi aadmi hai. Bina tracking ke na speed nikal sakti hai na direction.

Q2. Aapne konsa model train kiya?

YOLO humne train nahi kiya, pretrained weights use kiye hain. Jo model hum actually train karte hain woh anomaly detection ka Isolation Forest hai, aur woh har camera par uske apne data se live fit hota hai.

Q3. React mein AI chalta hai kya?

Nahi. React sirf visualization hai. Saara AI Python mein chalta hai. React FastAPI se JSON leta hai aur screen par dikhata hai.

Q4. Database kaunsa use kiya aur kyun?

SQLite. Kyunki woh file-based hai, alag server install nahi karna padta, aur border outpost par IT team nahi hoti. Central system banega to PostgreSQL par jaana hoga, aur hamari database layer alag hai isliye badalna asaan hai.

Q5. Ek machine par kitne camera chalenge?

Humne chaar camera ek hi machine par, bina GPU ke test kiye hain. Speed ke liye har doosra frame analyse hota hai aur pose sirf bade dikhne wale logon par chalta hai. Bade scale ka load test humne abhi nahi kiya.

Q6. Naye camera par model dobara train karna padega?

Nahi. Detection model general hai, aur hamara measurement body height par based hai, camera ke pixel par nahi. Isliye naya camera plug karo aur wahi rules lag jaate hain.

Q7. Alert kitni der mein aati hai?

Analysis har doosre frame par hoti hai aur dashboard har second poll karta hai. Matlab ghatna ke ek do second ke andar alert screen par hoti hai.

Q8. Evidence mein kya kya save hota hai?

Do JPEG: annotated poora frame aur us object ka close-up crop. ANPR ke liye plate crop alag. Video clip abhi save nahi hota, sirf images.

Q9. Aapka system offline chalega?

Haan. Poora AI local machine par chalta hai, internet ki zaroorat nahi. Cloud wala hissa abhi bana nahi hai, woh design mein hai.

Q10. Test kaise kiya?

491 automated test hain jo har build par chalte hain. Woh detection, tracking, behaviour, threat scoring, plate parsing aur login security sab cover karte hain. Video side par humne recorded border-jaisi footage use ki, kyunki asli border feed public nahi milti.

5 Difficult Questions

HARD 1. Isme naya kya hai? YOLO to aaj kal sab use karte hain.

> Sir, model naya nahi hai, model ka istemaal naya hai. Ek asli dikkat batata hoon jo humne solve ki.
>
> Har camera alag height aur alag angle par lagta hai, isliye ek hi aadmi ek camera mein bada aur doosre mein chhota dikhta hai. Agar speed pixel mein naapein to har camera ke liye alag calibration chahiye, jo hazaaron camera ke liye namumkin hai.
>
> Humne har cheez us aadmi ki apni body height ke hisaab se naapi hai. Chalna matlab 0.45 body height per second, daudna matlab 1.9. Yeh number har camera par same rehta hai. Nateeja: naye camera par koi calibration nahi chahiye. Stream jodo aur system chal padta hai.

HARD 2. Blockchain kahan hai? Theme to Blockchain aur Cybersecurity hai.

> Sir, main saaf bata deta hoon. Hamara kaam abhi cybersecurity wale hisse par hua hai, blockchain wale par nahi.
>
> Cybersecurity mein jo bana hai: password scrypt se hash hota hai, plain text mein kabhi nahi. Session token bhi seedha store nahi hota, uska fingerprint store hota hai. Do role hain aur permission backend mein check hoti hai, frontend mein nahi. Aur 18 tarah ke audit log bante hain.
>
> Blockchain abhi implement nahi kiya, aur main galat nahi bolna chahta. Lekin uski jagah saaf hai: har evidence file ka SHA-256 hash lekar ek chain banani hai jahan har record mein pichle record ka hash ho. Isse koi beech ka record badle to chain toot jaati hai. Yeh roadmap mein hai.

HARD 3. False alarm kitne aayenge? Jawaan pareshaan ho gaya to system band kar dega.

> Sir, yeh hamare liye sabse important sawaal hai, aur hum bhi yahi maante hain ki zyada false alarm system ko maar deta hai.
>
> Humne teen level ka bachaav rakha hai. Ek, koi track tab tak ginti mein nahi aata jab tak woh kam se kam 3 baar na dikhe, to ek frame ki jhalak kabhi alert nahi ban sakti. Do, boundary wali harkat pehle sirf arm hoti hai, alert tab jaati hai jab agle frames mein confirm ho. Teen, threat score ke 4 level hain aur LOW wali cheezein sirf log hoti hain, operator ko dikhti hi nahi.
>
> Aur jo alert jaati hai usmein wajah likhi hoti hai, to operator bina video dekhe hi samajh jaata hai ki dekhna kyun hai.

HARD 4. Raat mein kaam karega? Border ki nigraani to raat mein hi hoti hai.

> Sir, main iska seedha jawaab deta hoon. Abhi hamare paas low-light **measurement** hai, matlab system naap leta hai ki frame kitna andhera hai, aur usko threat score mein ek factor ki tarah use karta hai. Raat ka time apne aap mein risk badha deta hai.
>
> Lekin image enhancement ka code abhi nahi hai, aur main yeh nahi bolunga ki hai. Bilkul andhere mein normal camera ki apni physical limit hai.
>
> Iska asli hal thermal camera hai. Hamari strategy RGB aur thermal ko milane ki hai, aur architecture modular hai to woh module jud sakta hai. Par abhi woh bana nahi hai kyunki test ke liye thermal camera nahi tha.

HARD 5. Yeh market mein already nahi hai kya? Video analytics products to hain.

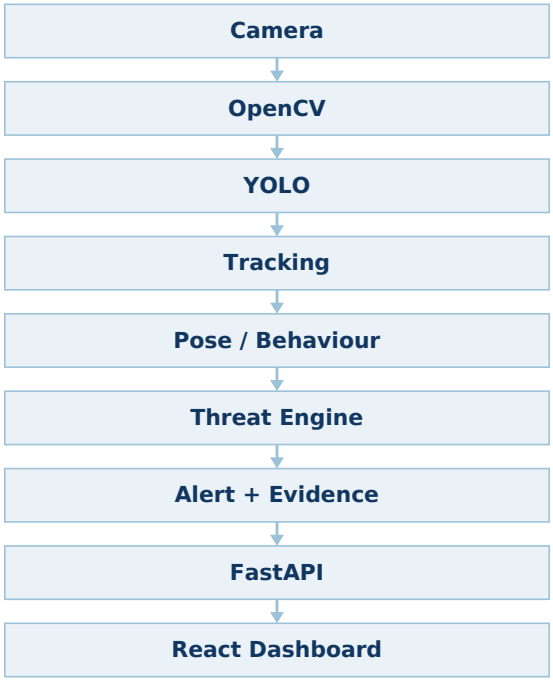
> Sir, products hain, lekin unmein do dikkat hain.
>
> Ek, zyadatar apna camera ya apna box bechte hain. Hamari poori shart yeh hai ki jo camera pehle se laga hai wohi chale, kyunki problem statement bhi yahi kehta hai.
>
> Do, woh general purpose hote hain, mall aur parking ke liye. Hamare rules border ke liye hain: fence ke paas rukna, ulti disha se crossing, raat ka time, group banana. Aur woh rules ek YAML file mein hain, to har sector apne hisaab se tune kar sakta hai bina naya software banaye.

SECTION 23

Quick Revision: Har Technology Ek Line Mein

OpenCV	Video ko frame mein todta hai aur frame ko AI ke layak banata hai. Result wapas frame par draw bhi yahi karta hai.
YOLO	Ek hi baar image dekhkar bata deta hai ki usmein kya hai aur kahan hai. Isliye live video par chal sakta hai.
ByteTrack	Har aadmi aur gaadi ko ek ID deta hai jo frame dar frame usi ke saath rehti hai. Isse speed aur direction nikalti hai.
Pose Estimation	Sharir par 17 point nikaalta hai, taaki pata chale ki aadmi khada hai, jhuka hai ya chadh raha hai.
Behaviour Analysis	Detection plus movement plus pose ko jodkar ek matlab wali activity banata hai, jaise RUNNING ya CROUCHING.
Threat Engine	Poora context dekhkar 0 se 100 ka score nikaalta hai aur LOW, MEDIUM, HIGH ya CRITICAL batata hai. Khatre ka faisla yahi karta hai, YOLO nahi.
Isolation Forest	Bina sikhaye anokhi harkat pakadta hai. Camera ka apna normal khud seekh leta hai.
FastAPI	Python AI aur React dashboard ke beech ka pul. Data ko JSON bana kar browser tak bhejta hai.
React	Dashboard ka visualization layer. Yahan AI nahi chalta, yeh sirf dikhata hai aur control deta hai.
SQLite	File-based database. Alag server ki zaroorat nahi, isliye border outpost ke liye sahi hai.
ANPR	Gaadi ka number plate apne aap padhna, aur us number se state aur district nikaalna.
OCR	Image mein likhe hue akshar ko text mein badalna. ANPR ke andar yahi kaam karta hai.
RTSP	Standard tareeka jisse camera apni live video network par bhejta hai. Har IP camera aur NVR yeh deta hai, isliye brand se farak nahi padta.
Edge Computing	AI wahin chalana jahan camera hai. Faayda: turant alert, kam bandwidth, aur internet na ho to bhi kaam chalta rahe.
Cloud	Central monitoring, lambe samay ka storage aur multi-site analytics. Hamare project mein yeh abhi design tak hai, bana nahi hai.

Final Architecture



AAKHRI BAAT

Aakhri baat. Judge ko perfect project se zyada woh team pasand aati hai jo apna project sach mein samajhti hai. Jo bana hai usko confidence se dikhao, jo nahi bana uska plan batao. Dono baatein ek hi saans mein bol sakoge to round accha jayega.